

060243123 Computer Network and Internet

เครือข่ายคอมพิวเตอร์และอินเทอร์เน็ต

Layer 4 – Transport Layer

- Introduction, Functions
- Transport protocols
- UDP, TCP



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TCP/IP Model - Functions

7	Application	User interface, application services (email, web browsing, file transfer)	Application	5
6	Presentation	Data formatting, encryption, compression		
5	Session	Session establishment, management, termination		
4	Transport	Reliable data delivery, flow control, congestion control	Transport	4
3	Network	Logical addressing (IP), routing	Network	3
2	Data Link	Physical addressing (MAC), error detection/correction, media access control	Data Link	2
1	Physical	Physical transmission of bits (cables, signals)	Physical	1

Transport Layer

- Is the 4th layer in the TCP/IP model, situated between the Application Layer and the Network Layer.
- It acts as a bridge, establishing logical communication channels between application processes running on different hosts across a network.
- Its task is to provide services to both the higher-level (Application Layer) and the lower-level (Network Layer).

Core Functions of the Transport Layer

- **Connection-Oriented Service (TCP)**

- Establishes and maintains reliable, persistent connections between application processes.

- **Segmentation and Reassembly (TCP)**

- A message is broken down into segments that can be transmitted. Each segment has a sequence number that allows the transport layer to reassemble the message.

- **Correct delivery order (TCP)**

- Ensures data arrives in the correct order and without errors before passing it to the upper layer.

- **Reliable transfer (TCP)**

- Guarantees that all data sent is received; if packets are lost, the protocol retransmits them.

Core Functions of the Transport Layer (Cont.)

- **Flow control (TCP/SCTP; not UDP)**
 - Limits sending based on receiver buffer capacity.
 - Congestion control (TCP/SCTP/QUIC; not basic UDP)
 - Adjusts sending based on network congestion signals.
- **Byte-stream service (TCP)**
 - Application data is delivered as an ordered byte stream; TCP does not preserve application message boundaries.
- **Multiplexing and demultiplexing (TCP/UDP/SCTP)**
- **Uses port numbers so many application processes can share the same host network stack.**

Physical and Logical Communications

Figure 1.18 *Communication through an internet*

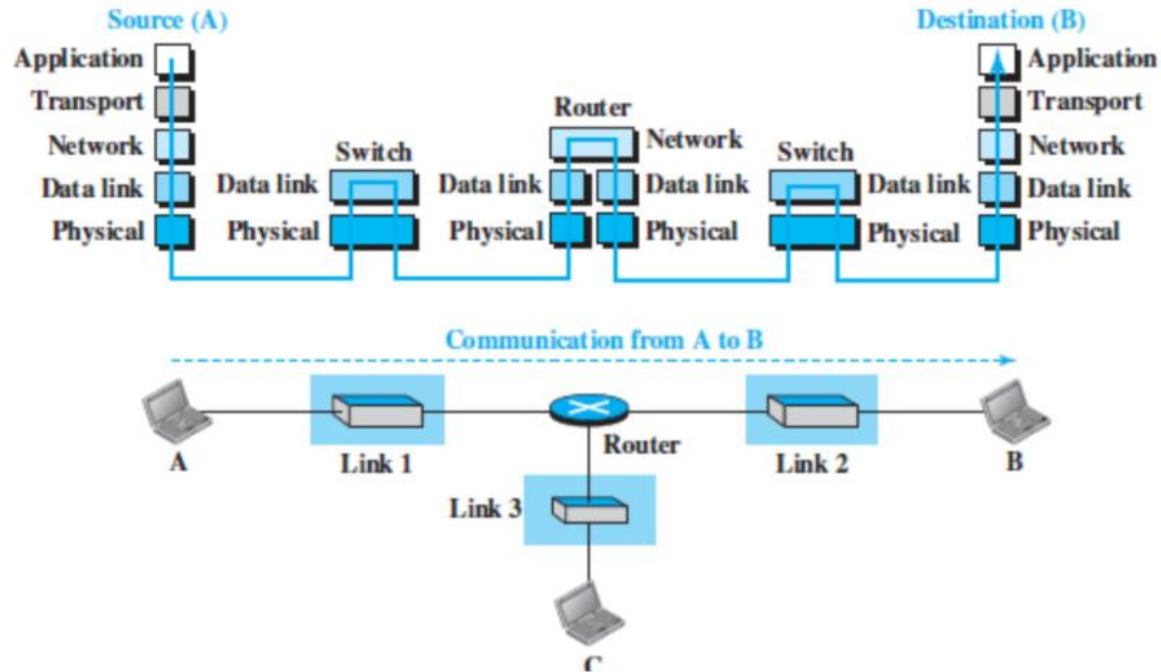
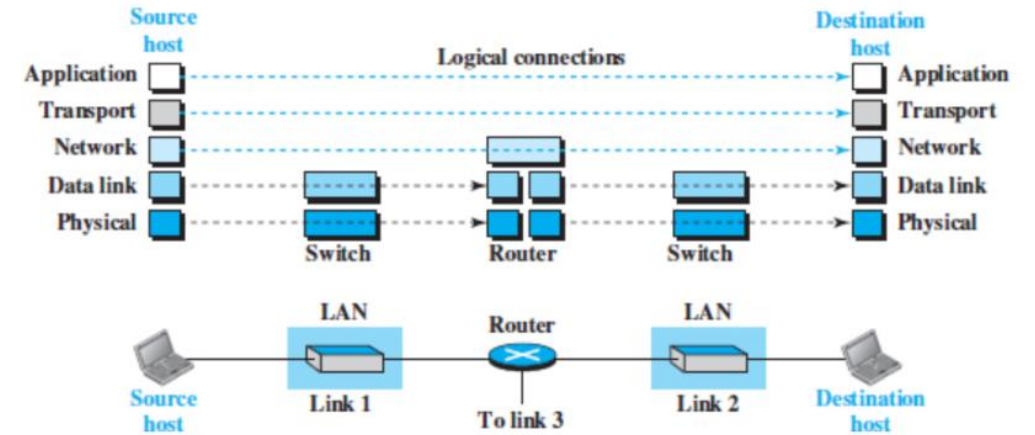
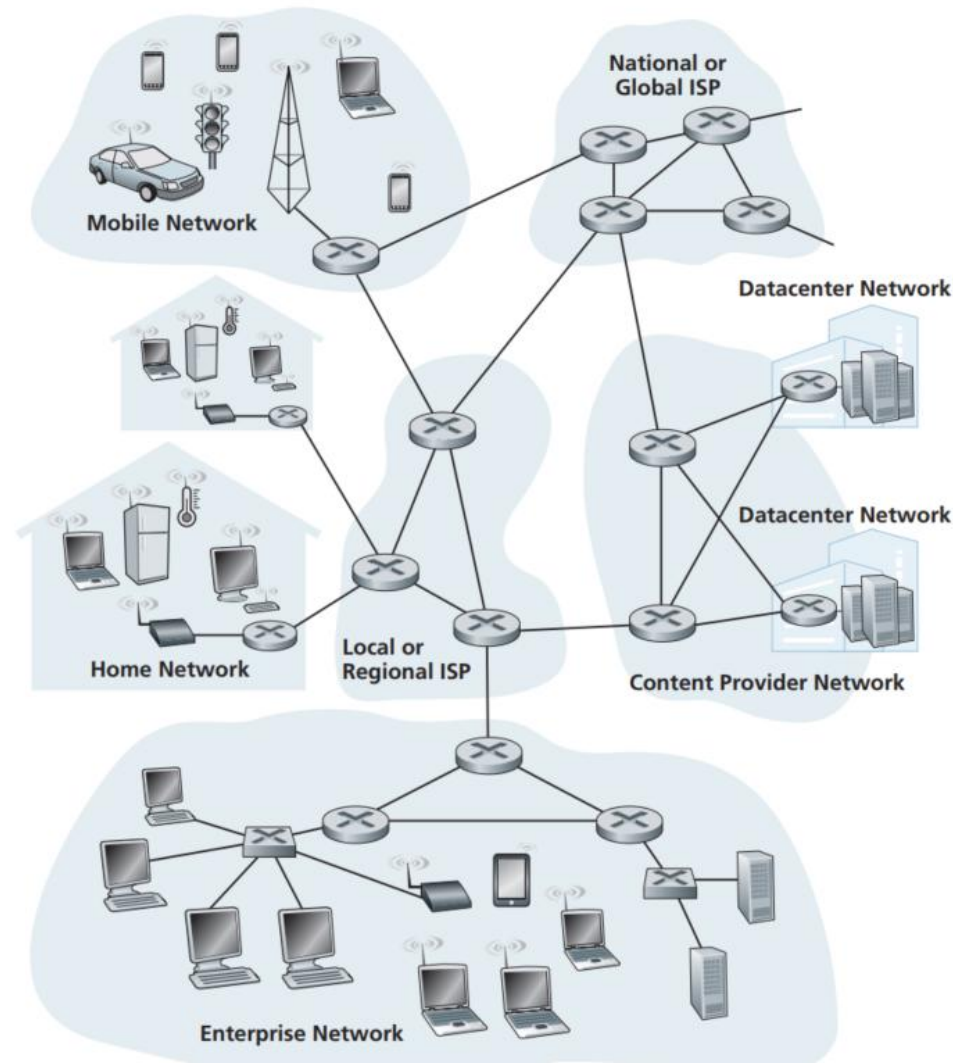


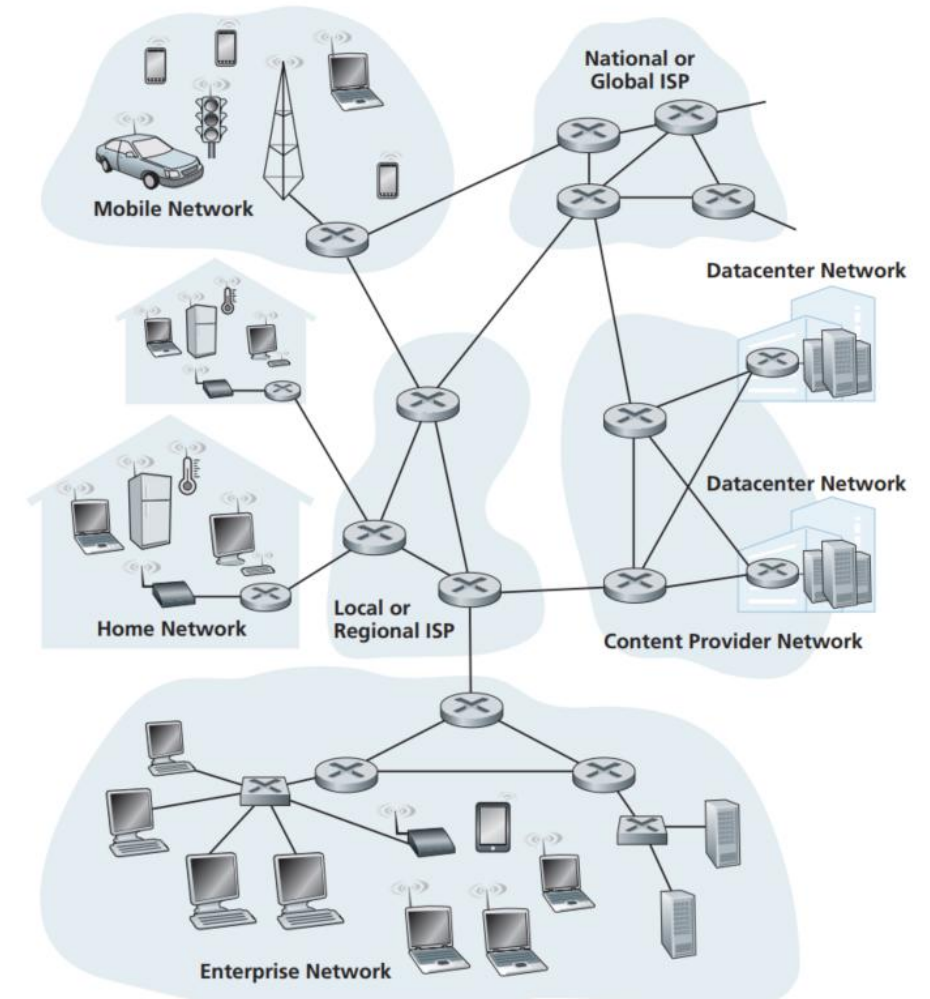
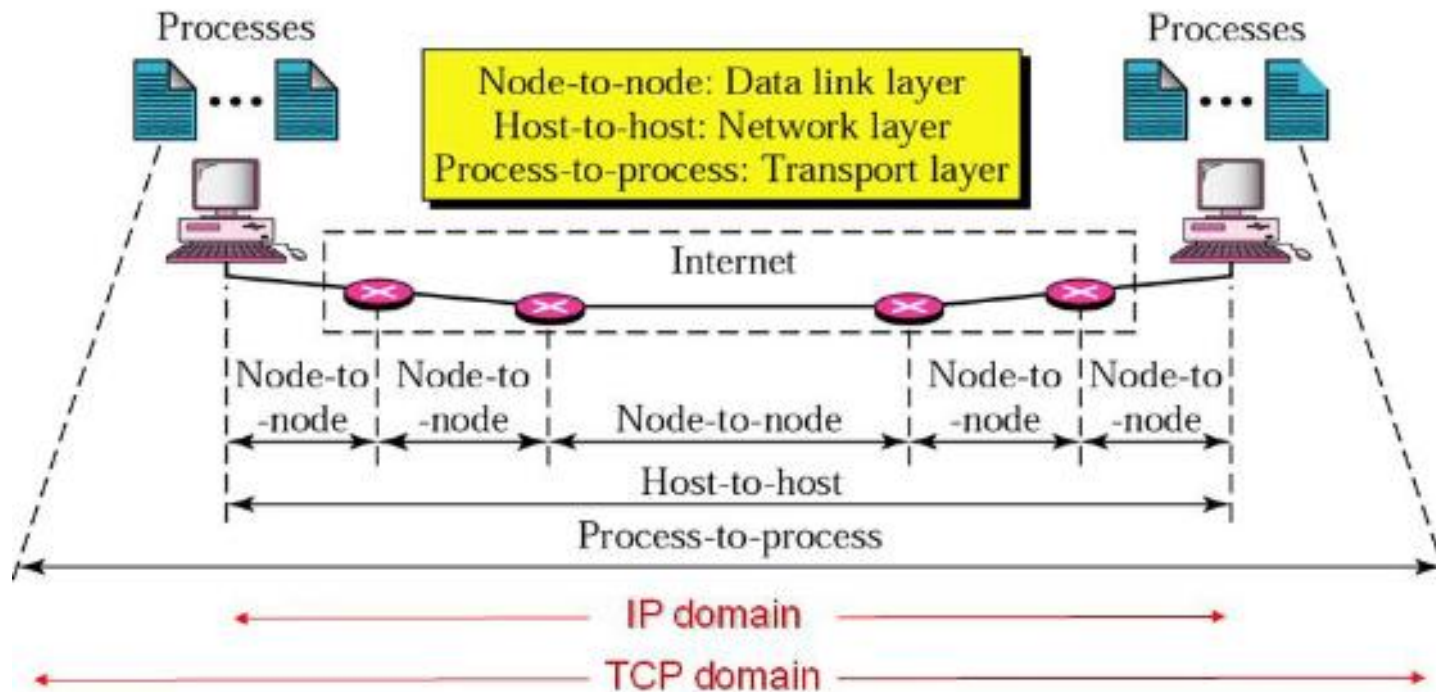
Figure 1.19 *Logical connections between layers of the TCP/IP protocol suite*



Types of Data Deliveries



Types of Data Deliveries



Data Deliveries in Transport Layer

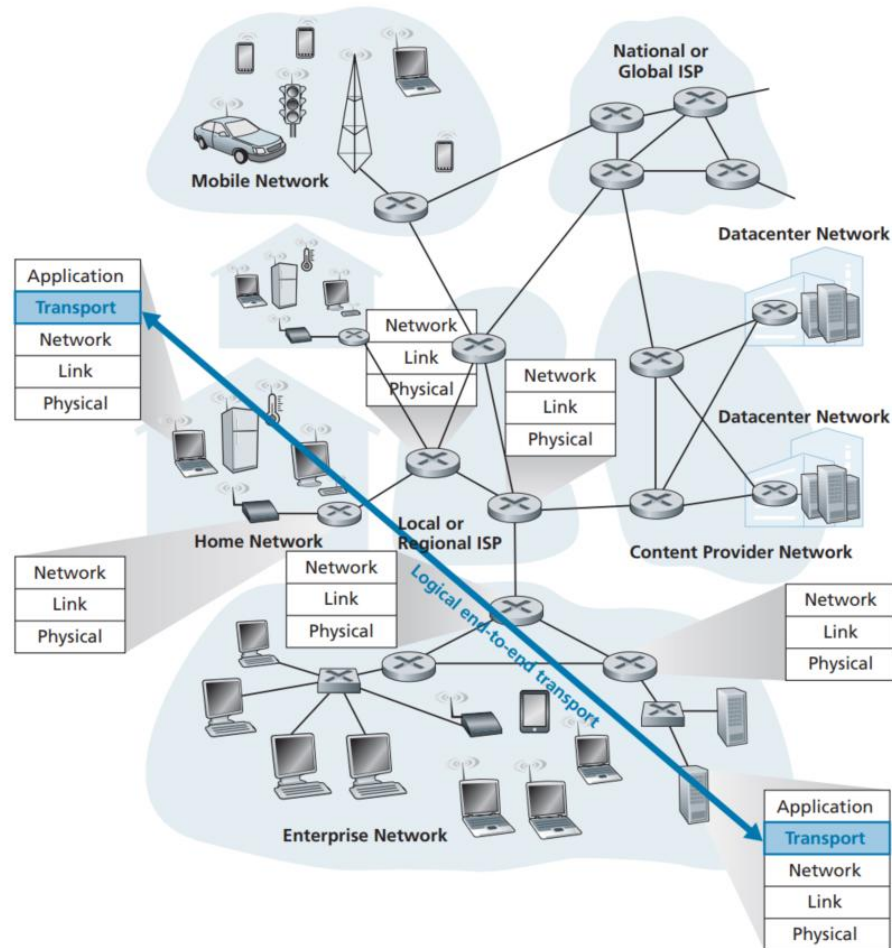
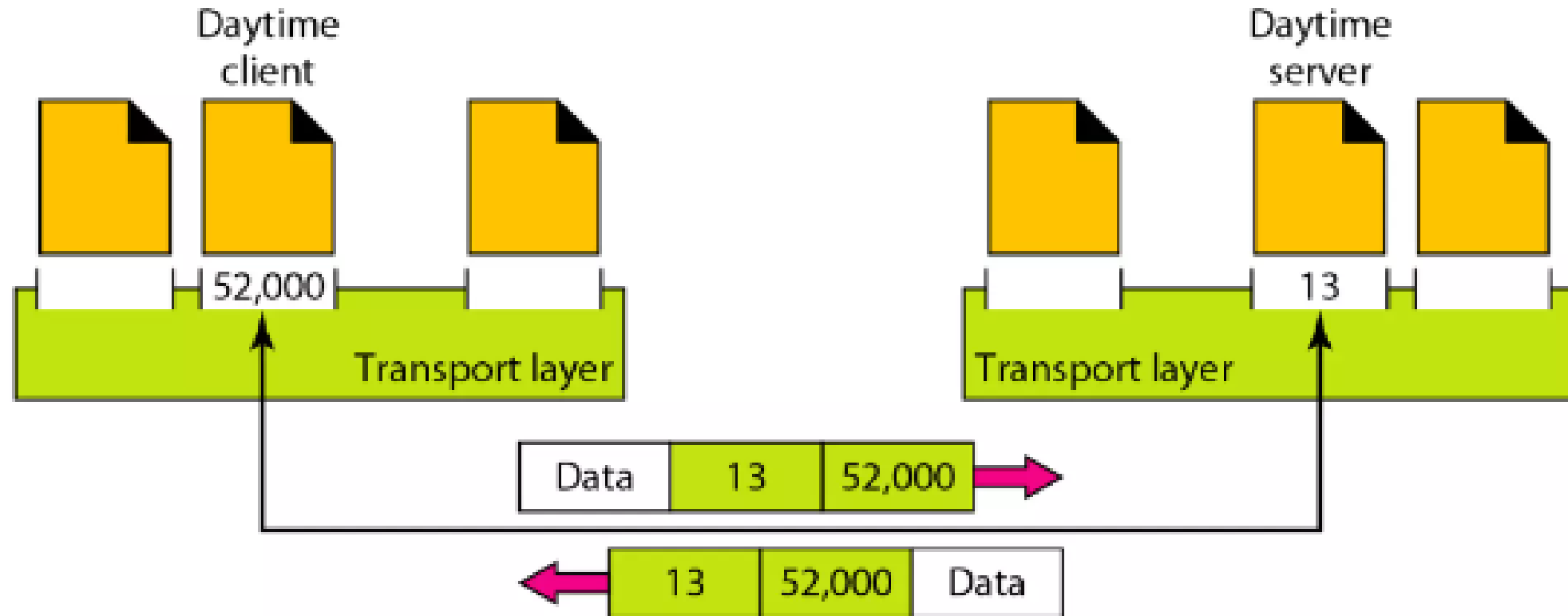
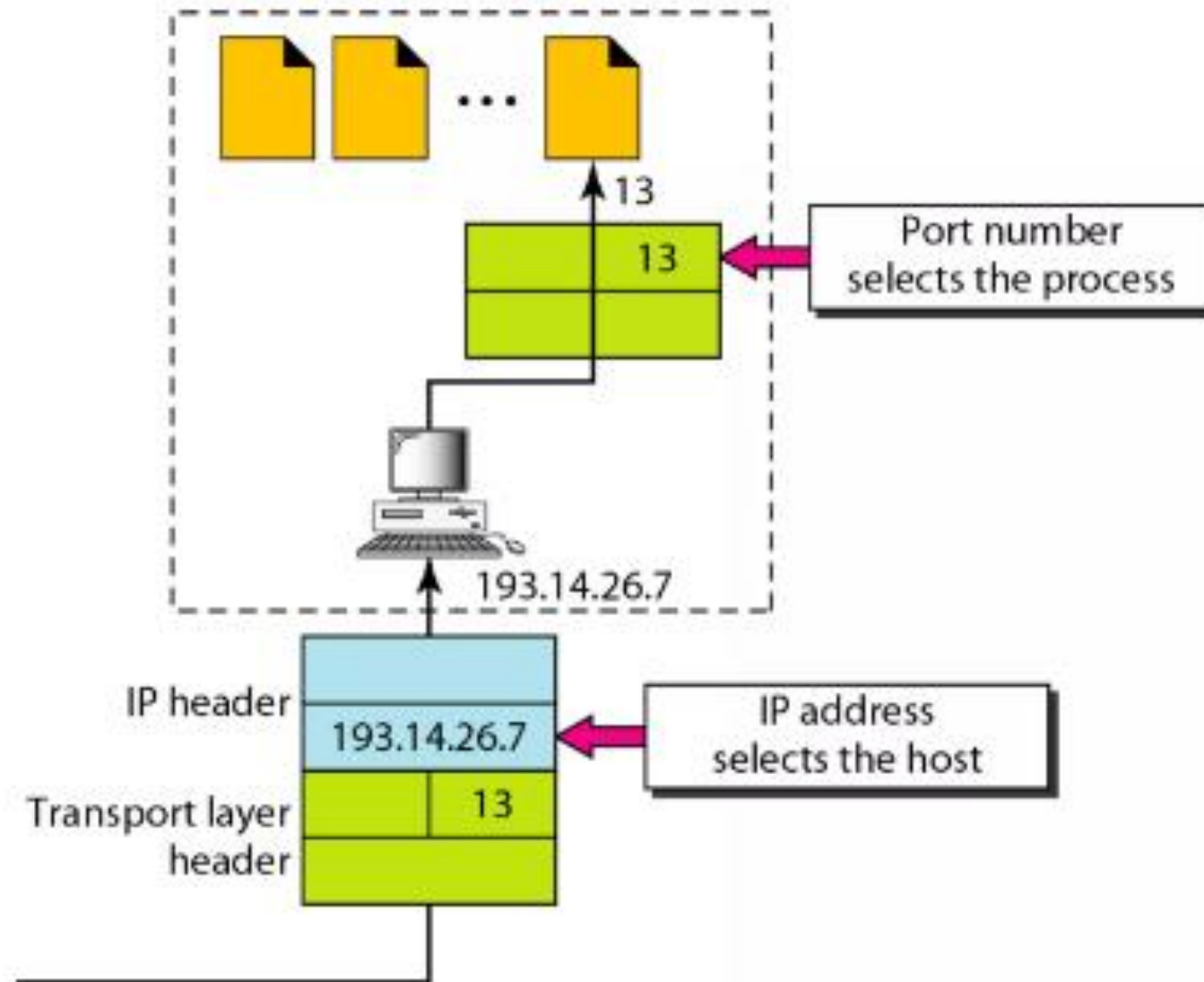


Figure 3.1 ♦ The transport layer provides logical rather than physical communication between application processes

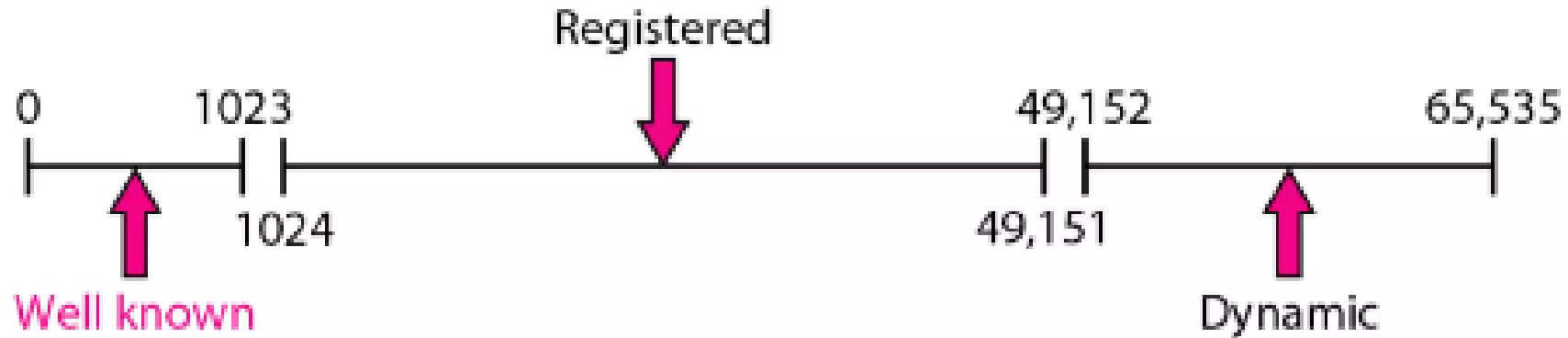
TRANSPORT ADDRESS



IP Address and Port Number



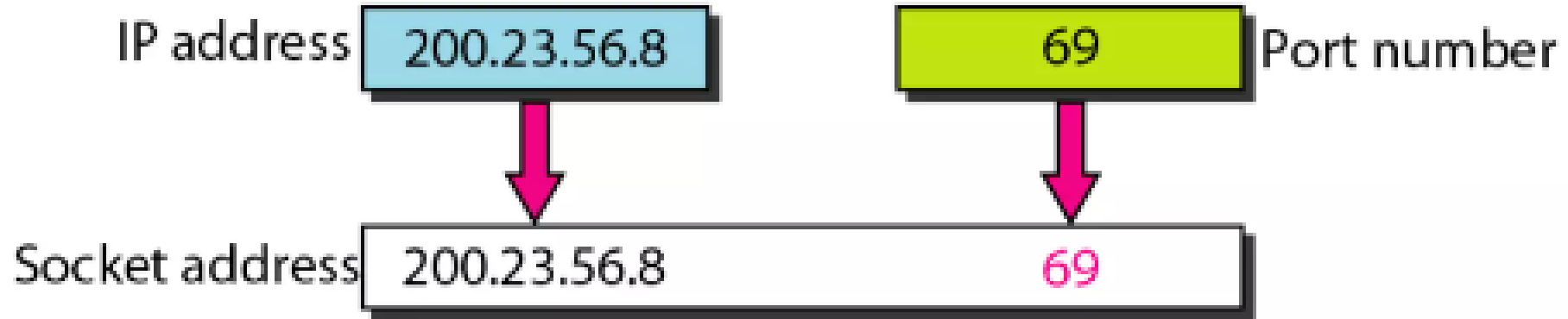
Port Number



IANA: Internet Assigned Number Authority

- **0 – 1023**
 - Well-known ports, managed by IANA
- **1024 – 49151**
 - Ports that can be registered with IANA
- **49152 – 65535**
 - Dynamic/Private ports (usually used for client ephemeral ports)

Socket Address



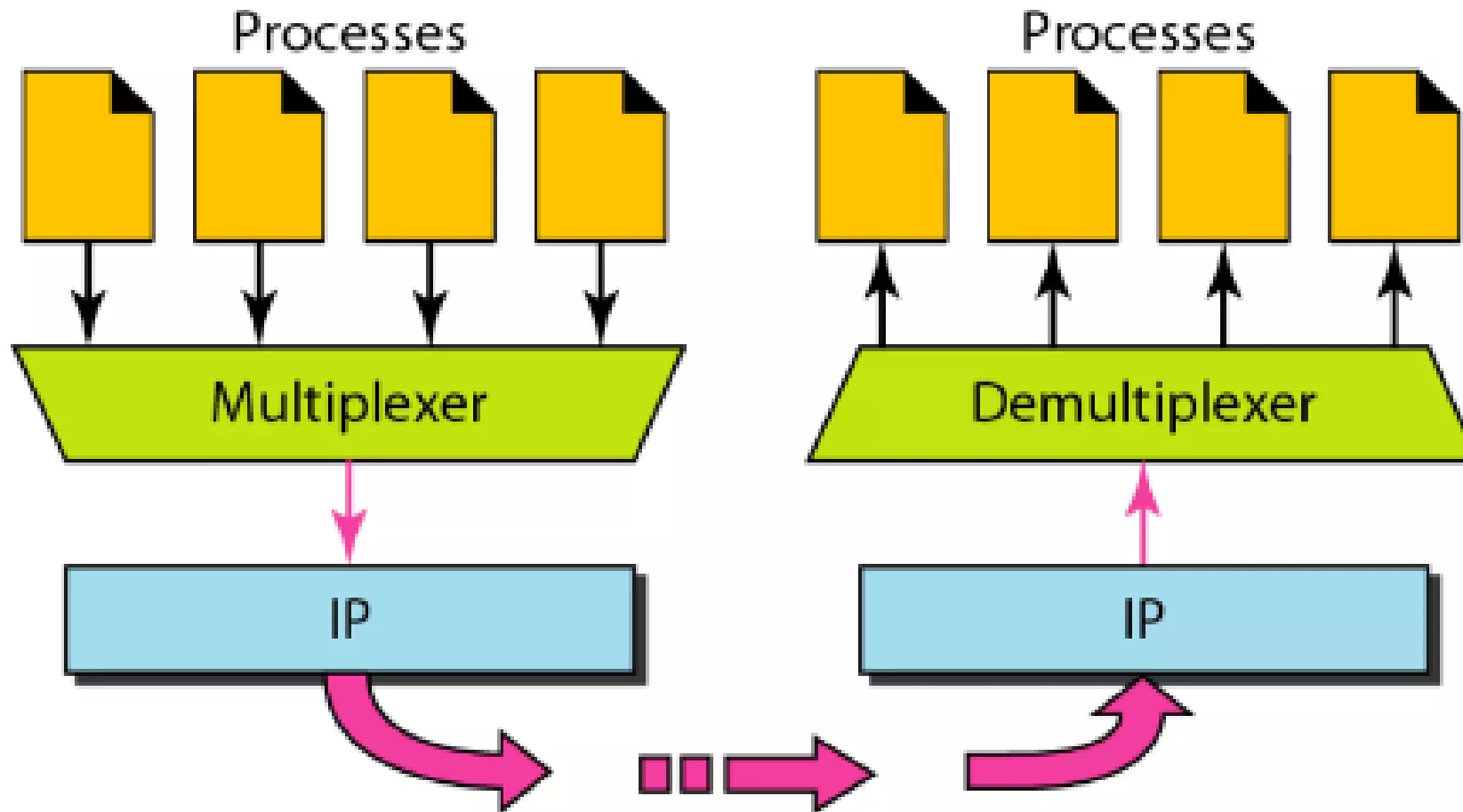
- **IP address:** A unique numerical label (32-bit in IPv4, 128-bit in IPv6) assigned to each device connected to a network.
- **Port number:** A 16-bit numerical identifier assigned to a specific process or service running on a device.

“The combination of an IP address and a port number identifies a communication endpoint on a host.”

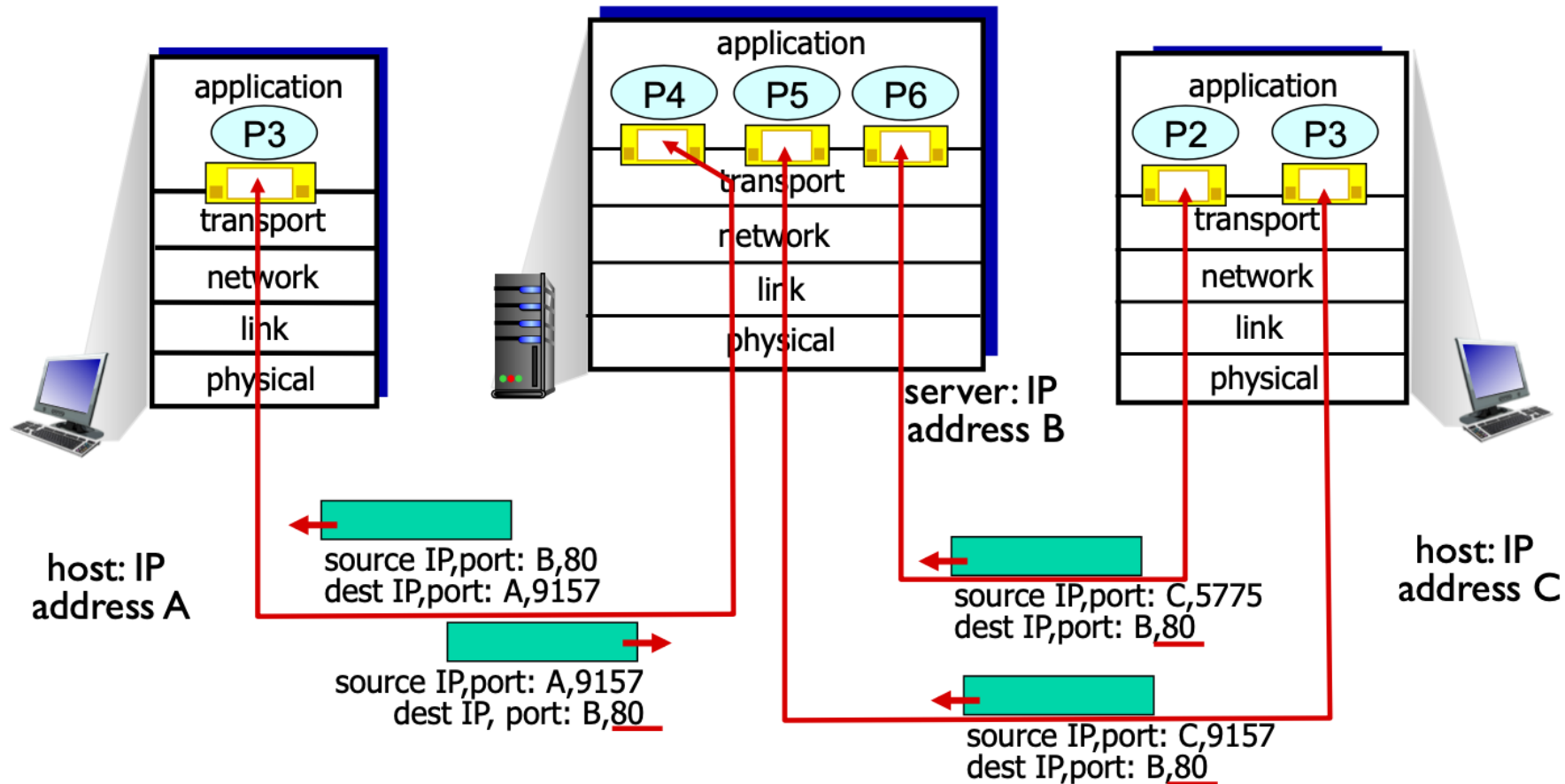
Some Well-Known Ports

Port	Protocol	UDP	TCP	SCTP	Description
7	Echo	√	√	√	Echoes back a received datagram
9	Discard	√	√	√	Discards any datagram that is received
11	Users	√	√	√	Active users
13	Daytime	√	√	√	Returns the date and the time
17	Quote	√	√	√	Returns a quote of the day
19	Chargen	√	√	√	Returns a string of characters
20	FTP-data		√	√	File Transfer Protocol
21	FTP-21		√	√	File Transfer Protocol
23	TELNET		√	√	Terminal Network
25	SMTP		√	√	Simple Mail Transfer Protocol
53	DNS	√	√	√	Domain Name System
67	DHCP	√	√	√	Dynamic Host Configuration Protocol
69	TFTP	√	√	√	Trivial File Transfer Protocol
80	HTTP		√	√	Hypertext Transfer Protocol
111	RPC	√	√	√	Remote Procedure Call
123	NTP	√	√	√	Network Time Protocol
161	SNMP-server	√			Simple Network Management Protocol
162	SNMP-client	√			Simple Network Management Protocol

Multiplexing and Demultiplexing



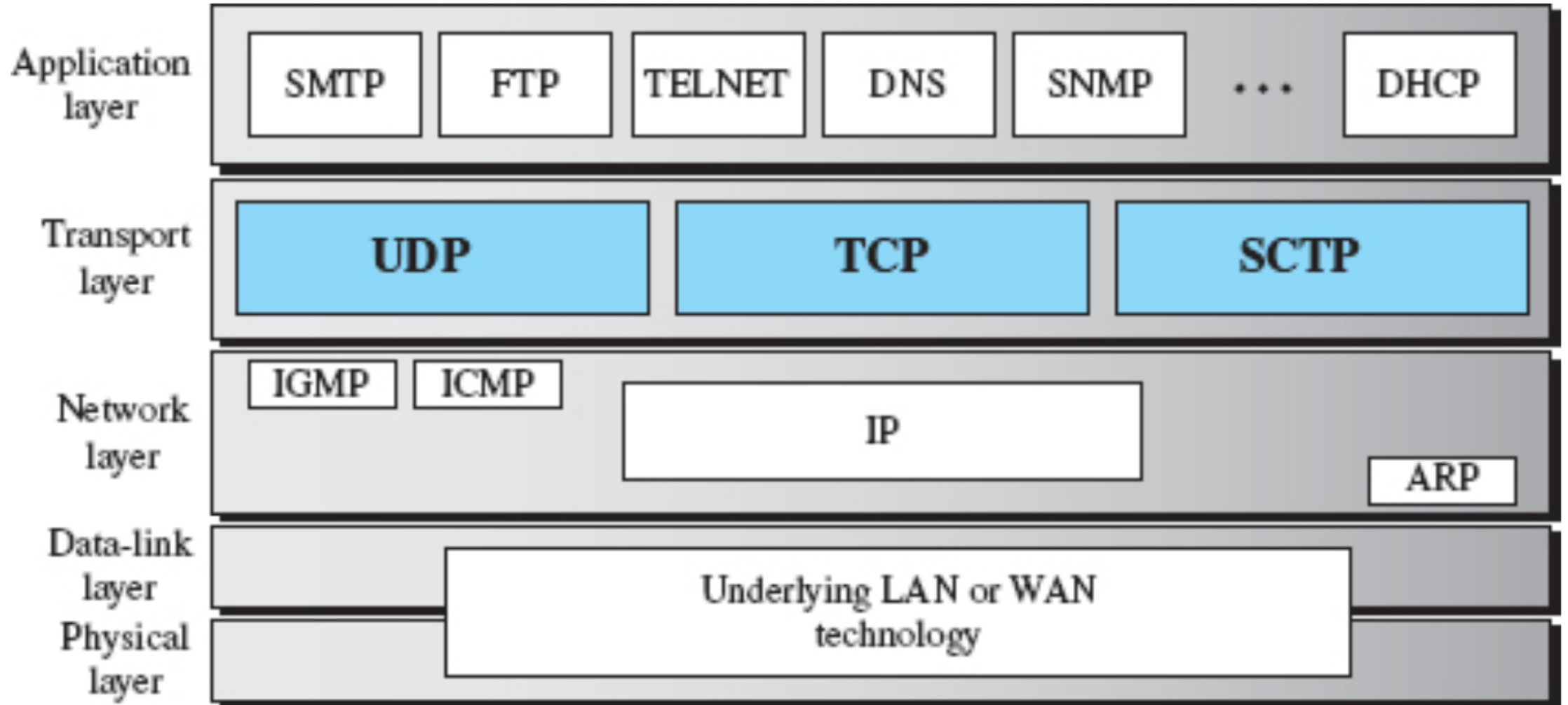
Multiplexing and Demultiplexing



three segments, all destined to IP address: B,
dest port: 80 are demultiplexed to *different* sockets

Transport Layer 3-12

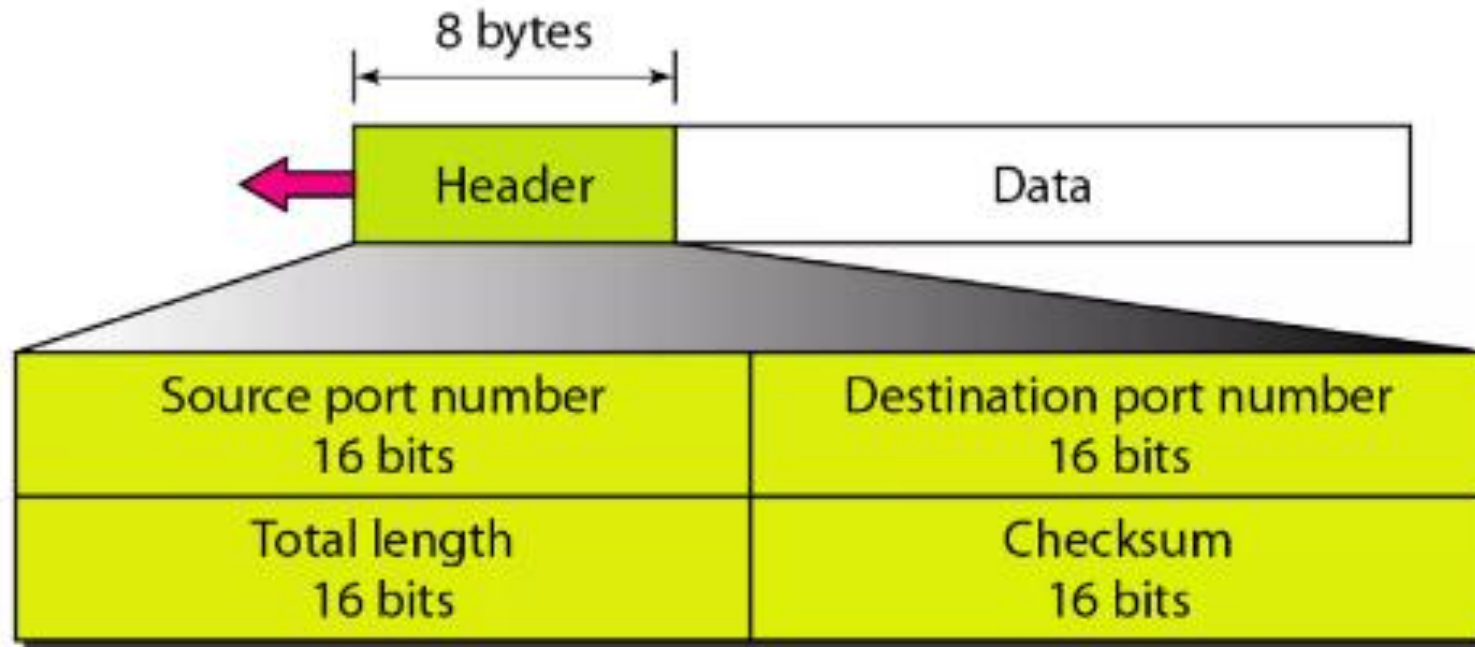
Protocols (UDP, TCP and SCTP) on Transport Layer



UDP – User Datagram Protocol

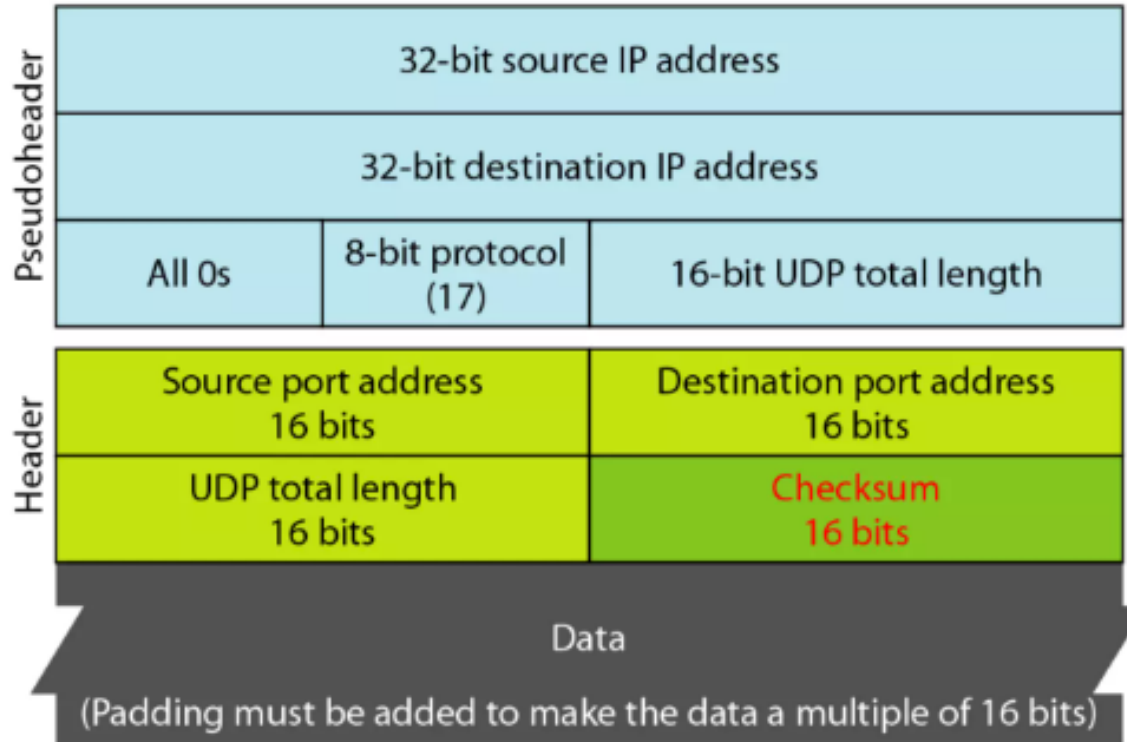
- **Connectionless:** no handshake; datagrams are sent independently.
- **Unreliable service:** no guarantee of delivery, order, or duplicate protection.
- **Lightweight:** small 8-byte header and low overhead.
- **Process-to-process delivery** using port numbers.
- **No built-in flow control or retransmission.**
- **Checksum provides error detection only; applications add reliability/congestion control if needed.**

UDP Packet Format



- UDP packets, called **user datagrams** have a fixed-size header of 8 bytes made up of four fields, each of 2 bytes (16 bits)

Checksum in UDP Header



- A checksum is a 16-bit field used for error detection.

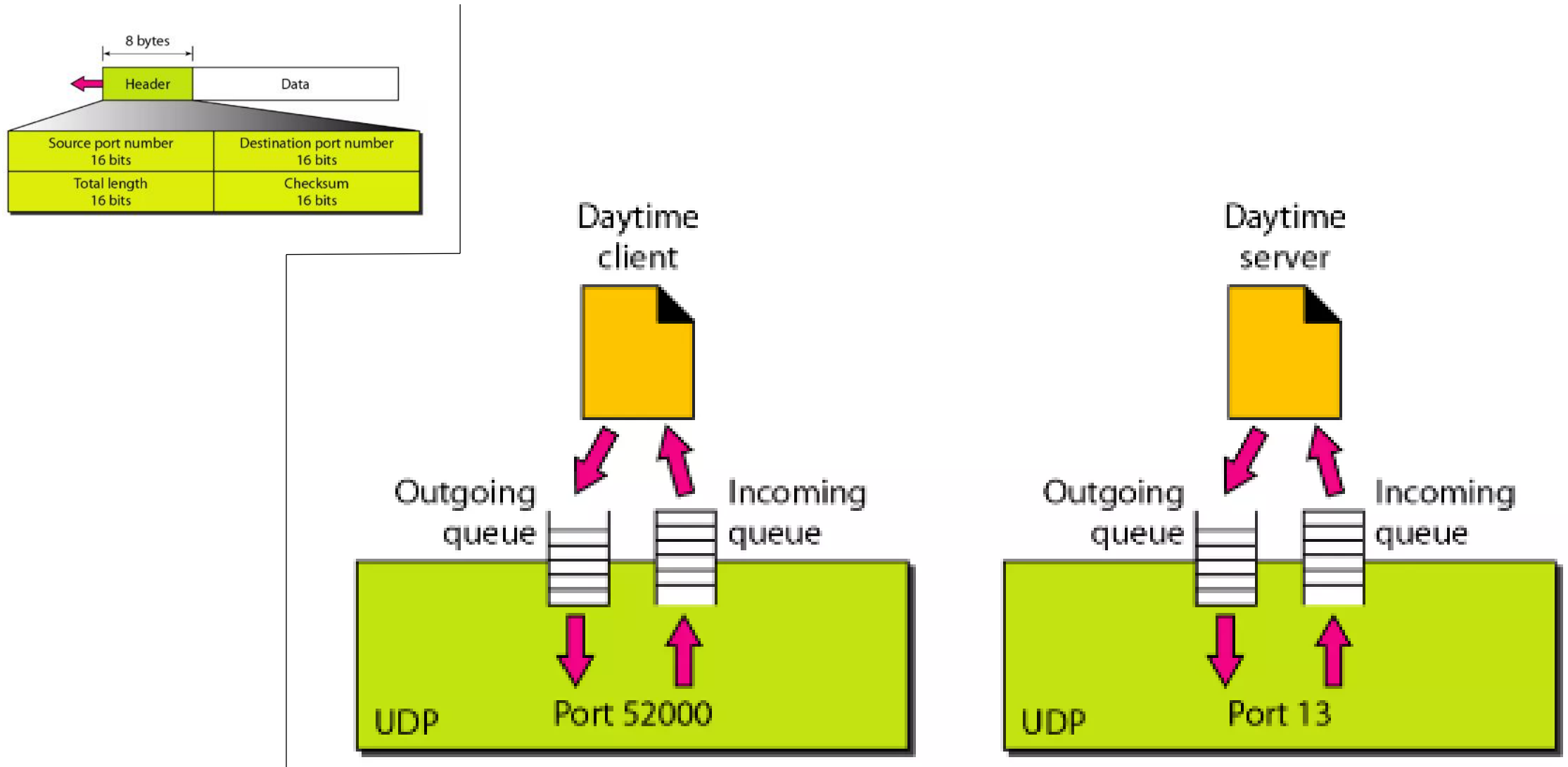
Sender computes it over the UDP pseudo-header, UDP header, and data.

Receiver recomputes and compares the checksum.

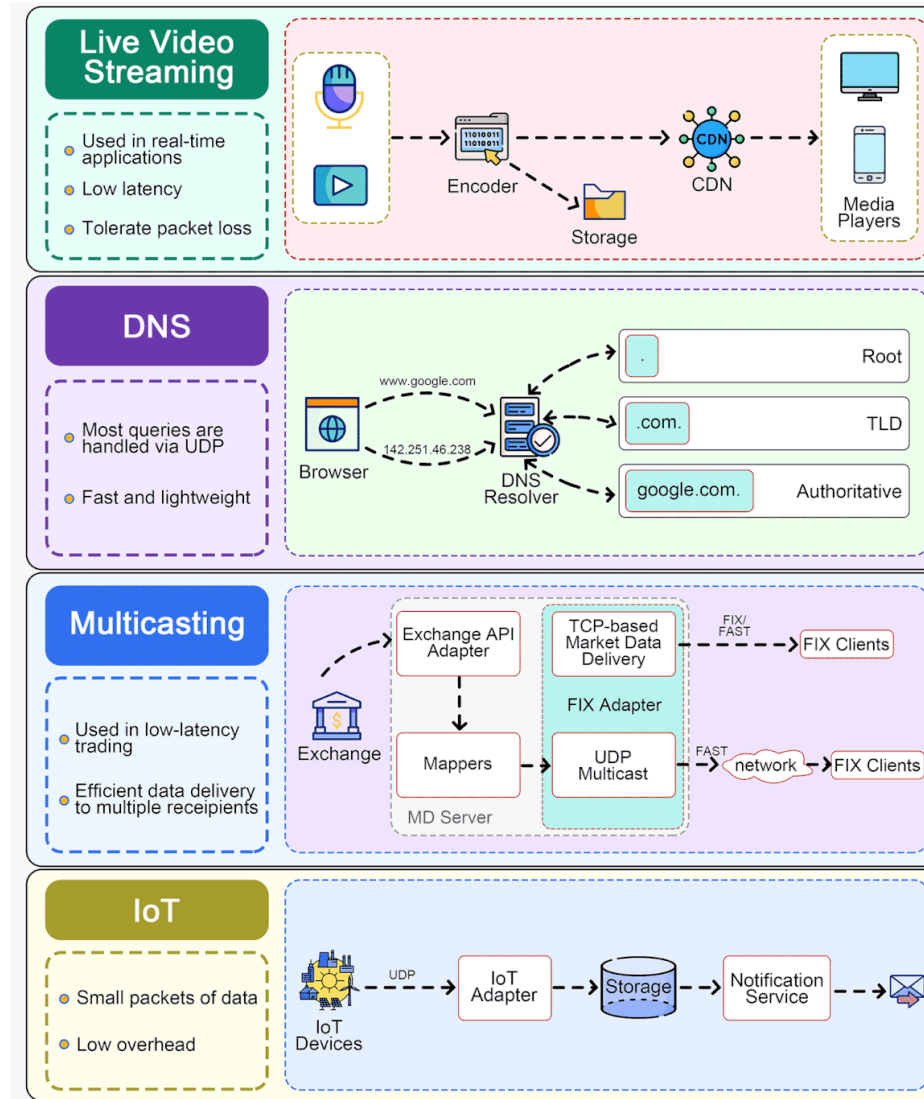
If it does not match, the datagram is discarded.

UDP checksum is optional in IPv4 and mandatory in IPv6.

Queues in UDP



Examples UDP Use Cases



Workshop: Data Encapsulation – Transport Layer

Draw Transport Layer Header of UDP from the top 2 layers, show encapsulation into Transport Layer headers.

DNS over UDP (Client → Server) เช่น DNS

- Source Port: 5000
- Destination Port: 53
- Data: DNS request